8 (a) (i) *collision rate:* molecule takes more time to complete one round trip and return to hit same wall B1

leads to decrease in collision rate B1

gas pressure: (from eq. 8.2) smaller force due to lower collision rate (and change in momentum per collision remains constant) B1

so pressure (= force / area) decreases B1

(ii) *collision rate:* (effective free space inside the container is reduced when volume of the gas molecules is taken into account,)

molecule travels less distance to complete one round trip and return to hit same wall B1

leads to increase in collision rate B1

gas pressure: (from eq. 8.2) larger force due to higher collision rate (and change in momentum per collision remains constant) B1

so pressure (= force / area) increases B1

(b) (i) calculate product of *p* and *V* of any two points e.g. (1.0, 3.2), (1.44, 2.0), (1.5, 1.9), (1.8, 1.5), (1.9, 1.4) M1

valid conclusion e.g. since product of *p* and *V* are not equal, temperature of process is not constant A1

(ii) average or mean kinetic energy (not internal energy) proportional to thermodynamic temperature B1

gas molecules transfer momentum to the piston during collision / molecules rebound from the piston with smaller momentum M1

average or mean (translational) kinetic energy of the gas molecules decreases A1

so, temperature decreases A0

(iii) 1. substitute in pV^{β} any two points from graph [e.g. (1.0, 3.2), (1.44, 2.0), (1.5, 1.9), (1.8, 1.5), (1.9, 1.4)] and equate C1

correct manipulation to obtain = 1.29 or 1.3 A1

$p_1V_1^{\beta} = p_2V_2^{\beta}$ $\quad \frac{V_1^{\beta}}{V_2^{\beta}} = \frac{p_2}{p_1}$ $\quad \left(\frac{V_1}{V_2}\right)^{\beta} = \left(\frac{p_2}{p_1}\right)$ $\quad \beta\ ln\ ln\left(\frac{V_1}{V_2}\right) = ln\ ln\left(\frac{p_2}{p_1}\right)$

$$\beta = \frac{ln\ ln\left(\frac{p_2}{p_1}\right)}{ln\ ln\left(\frac{V_1}{V_2}\right)} = \frac{ln\ ln\left(\frac{1.4}{3.2}\right)}{ln\ ln\left(\frac{1.0}{1.9}\right)}$$

2. no unit / dimensionless B1

(iv) calculate area under graph to obtain work done on gas between 1700 J and 1800 J A2
(if value 1.6 – 1.7 kJ OR 1.8 – 1.9 kJ, award one mark only)

Zero mark if values outside range.
Deduct one mark if answer includes negative sign.

V / 10^{-3} m^3	1.0	1.22	1.28	1.4	1.64	1.9
p / MPa	3.2	2.3	2.1	1.8	1.5	1.4
area or trapezium / 10^3 J	–	0.605	0.132	0.234	0.396	0.377

work done on gas = total area = 1.74 kJ
($|W_B| < |W_A|$. For comparison, work done by process A = 1.89×10^3 J.)

(v) [Thinking process:
- Apply first law to the cyclical process

process	increase internal energy	heat supplied	work done on gas
A	negative	0	negative
B	positive	positive	positive
net change of quantity	0	positive	negative (area of A > B)

- net work done by gas is positive
- heat supplied to gas in process B

]

net work done by gas = heat supplied (during process B) M1

resulting in 100% efficiency, hence not possible A1